ECML PKDD 2024

MT-HCCAR: Multi-Task Deep Learning with Hierarchical Classification and Attention-based Regression for Cloud Property Retrieval

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UMBC

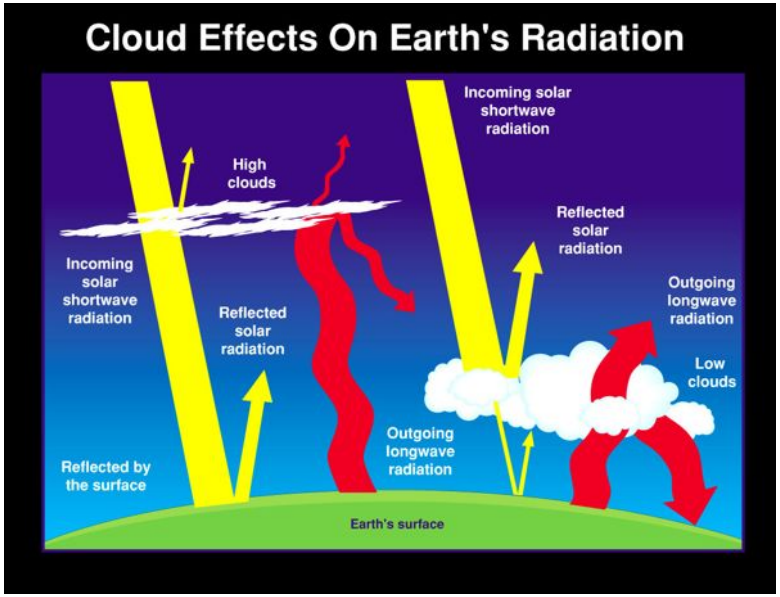
NASA Goddard SPACE FLIGHT CENTER

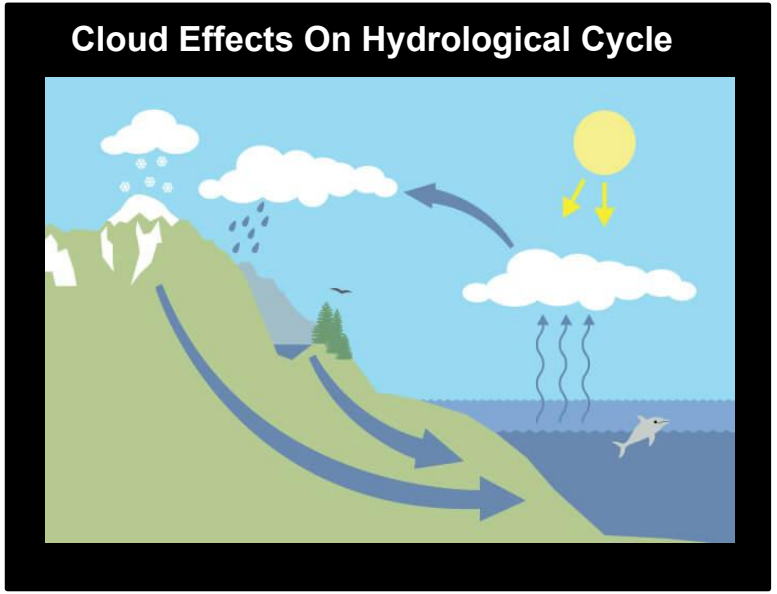
GESTAR II

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APPLICATION BACKGROUND

Clouds play a key role in Earth's atmospheric system.



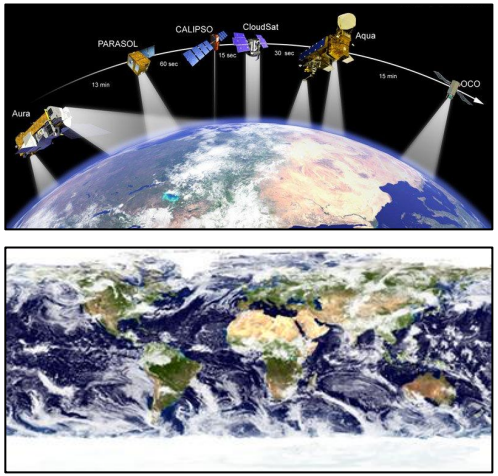


APPLICATION BACKGROUND

Cloud Property Retrieval

- The estimation of diverse characteristics of clouds through the analysis of remote sensing data.

Remote Sensing Data



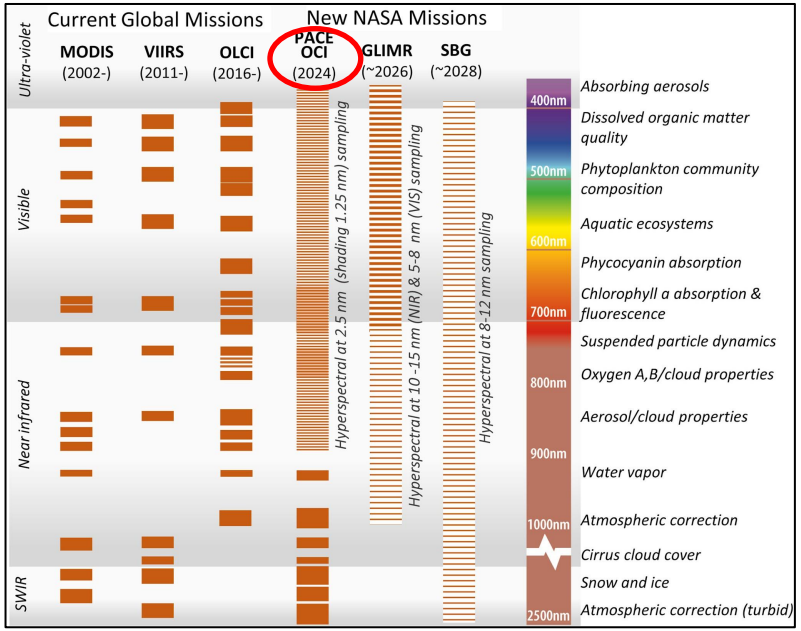
Cloud Properties

- Cloud Mask (e.g., a satellite pixel is cloudy or clear)
- Cloud Phase (e.g., a cloudy pixel is liquid or ice)
- Cloud Vertical Distribution (e.g., cloud top height)
- Cloud Optical Thickness (COT)
- Cloud Droplet Effective Radius (CER)
- ...

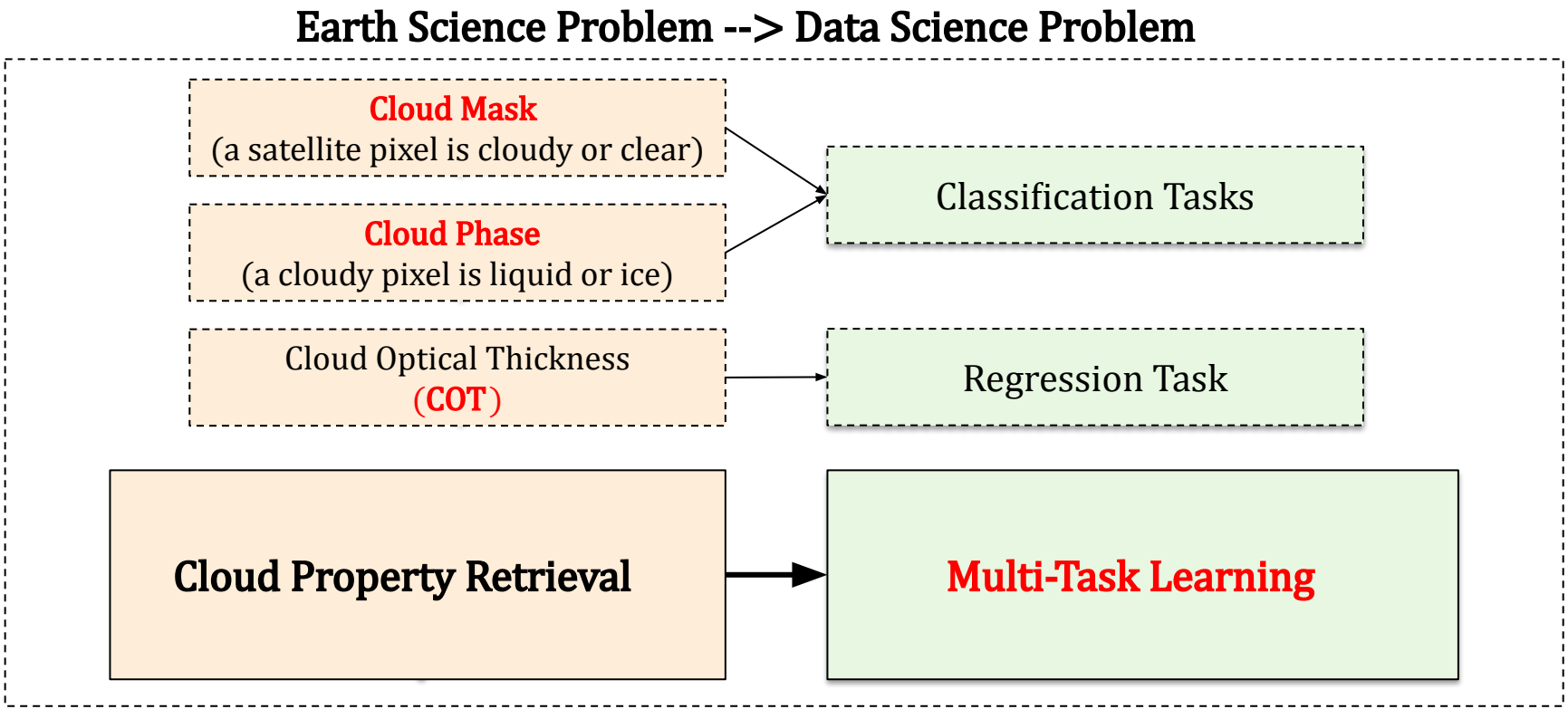
APPLICATION BACKGROUND

Our remote sensing data: Earth's Imagery

- **Reflectances** : observations of reflected solar or emitted thermal radiation across multiple wavelength ranges.
- Our practical motivation is **OCI**, the main sensor of Plankton, Aerosol, Cloud, ocean Ecosystem (PACE) mission.



INTRODUCTION



CONTRIBUTIONS

- **Model architecture** : We propose an end-to-end multi-task learning model **MT-HCCAR** for cloud masking, cloud phase classification, and COT regression.
- **Implementation** : We implement and evaluate models on three satellite sensors with different ranges of wavelength coverage (**OCI**, **VIIRS**, and **ABI**)
- **Model selection** : To help identify the best model for practical usage, we design a model selection method that ranks the performance of different models across multiple metrics and multiple datasets.

PROBLEM STATEMENT

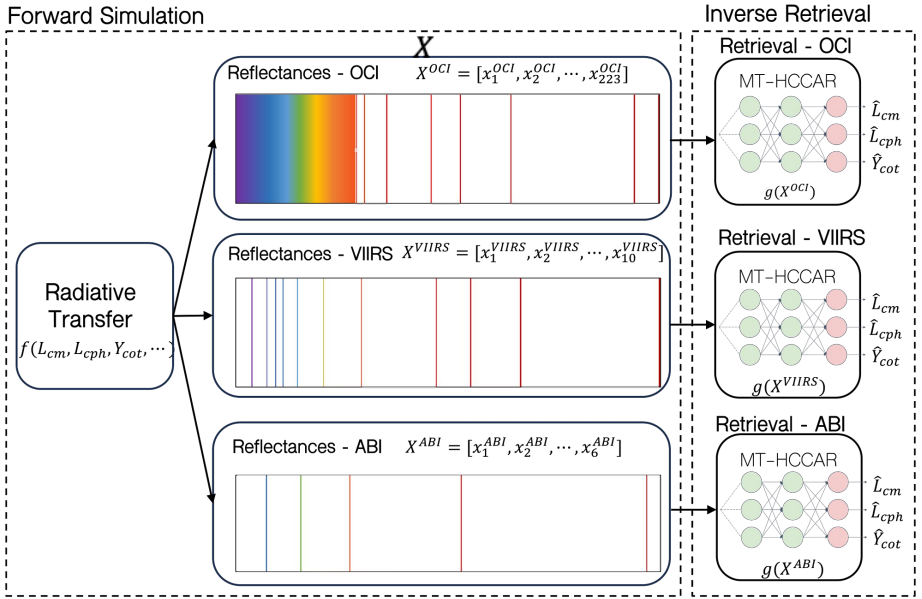
Forward simulation and **inverse retrieval** : two opposite operations within the context of cloud property analysis.

- Forward simulation: Data simulation

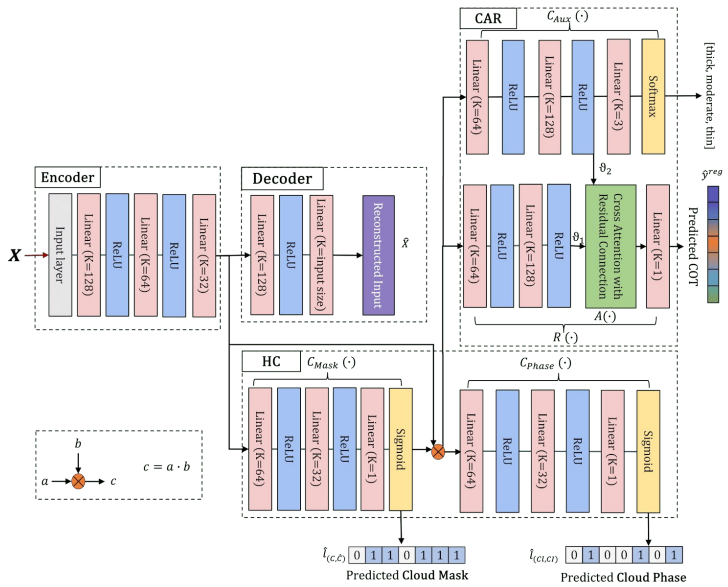
$$X = f(y_{cloud_{mask}}, y_{cloud_{phase}}, y_{COT}, \dots)$$

- Inverse retrieval: Cloud property retrieval

$$\hat{y}_{cloud_mask}, \hat{y}_{cloud\ phase}, \hat{y}_{COT} = g(X)$$



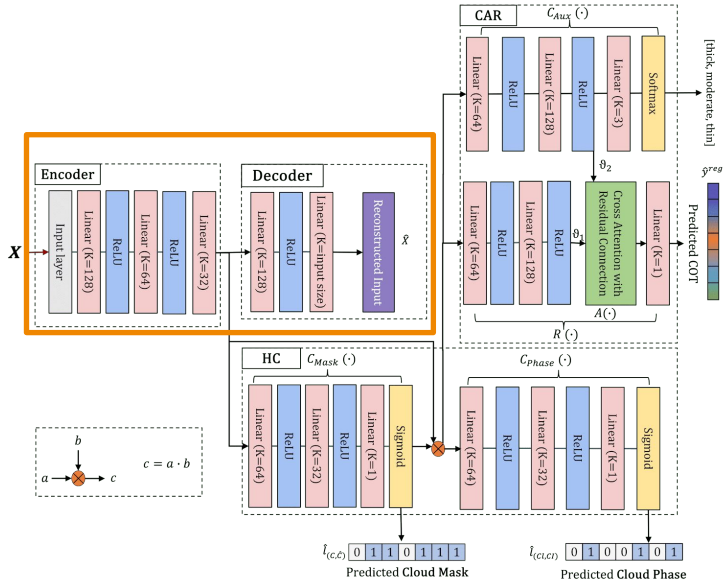
MT-HCCAR MODEL



Sub-networks:

1. **Encoder-Decoder**
2. **Hierarchical classification (HC)**
3. **Classification-assisted and Attention-assisted Regression sub-network (CAR)**

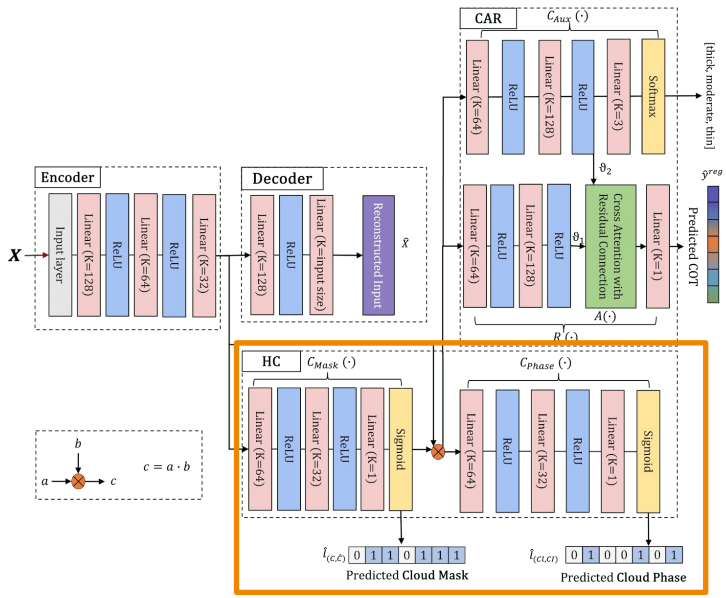
MT-HCCAR MODEL - 1



1. Encoder-Decoder

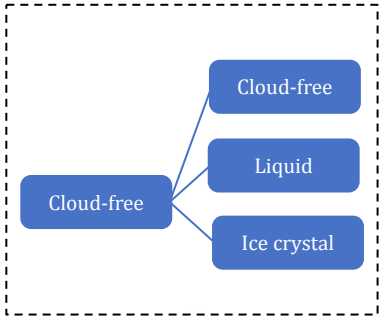
- An encoder-decoder sub-network contains an encoder module and a decoder module.
- Learns shared latent features for all tasks.
- Output is reconstruction of *X*.

MT-HCCAR MODEL - 2

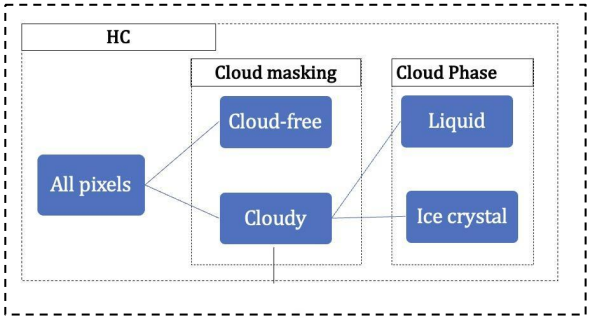


2. Hierarchical Classification (HC)

- Estimates cloud mask and cloud phase in a hierarchical way
- Categorizes pixels according to the level of labels.

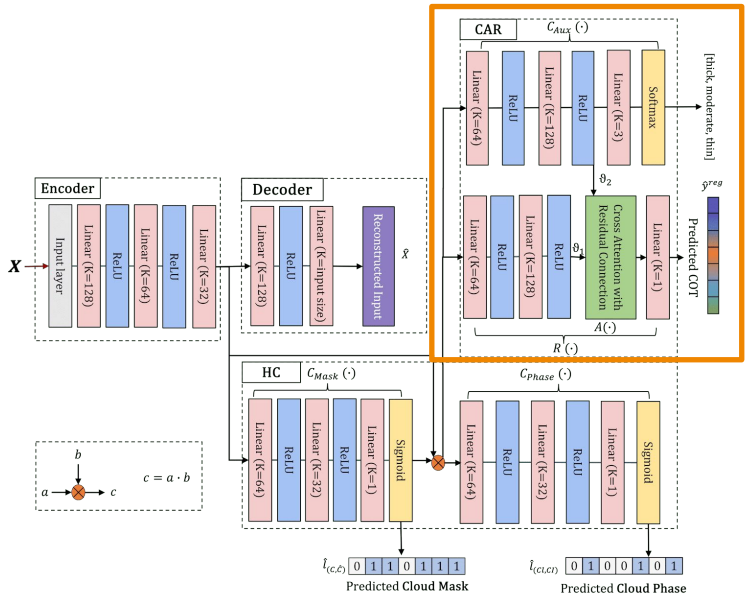


Multi-label classification



Hierarchical classification

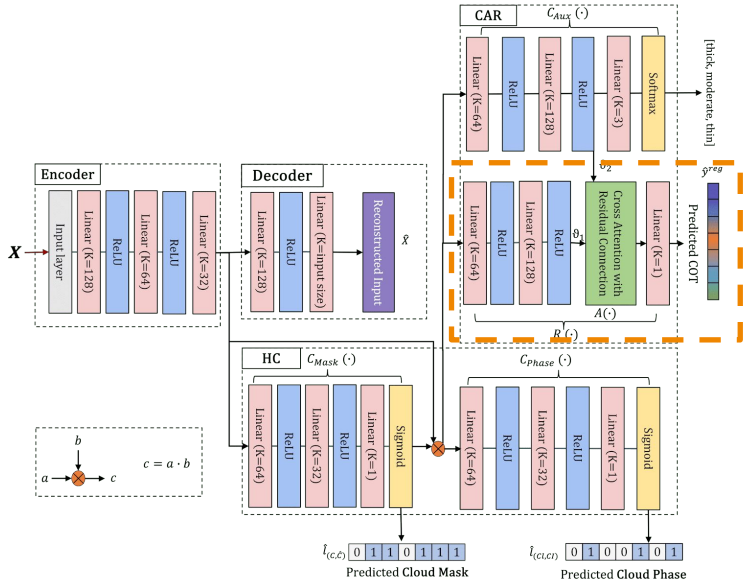
MT-HCCAR MODEL - 3



3. Classification-assisted and Attention-assisted Regression sub-network (CAR)

- auxiliary classifier (C)
- cross-attention module (A)
- regression network (R)

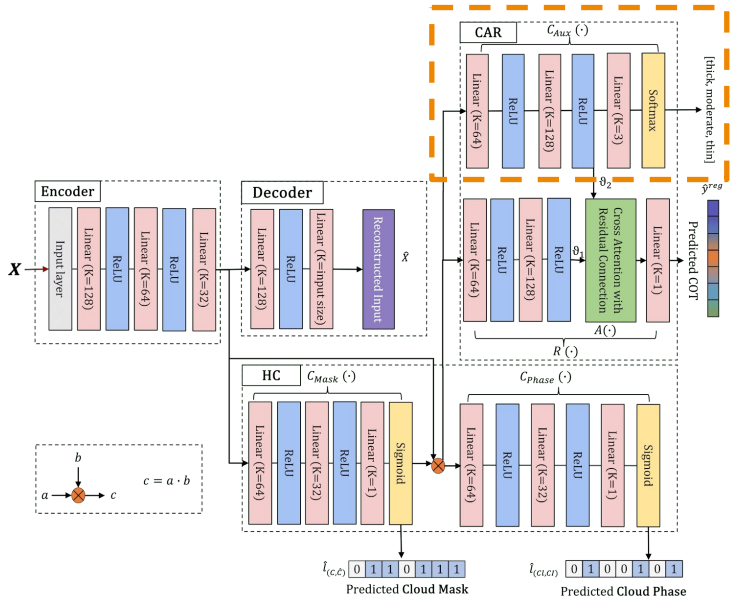
MT-HCCAR MODEL – 3.1



3.1 Regression network (R)

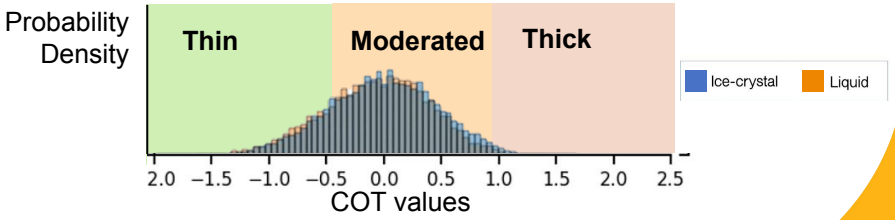
- Make predictions of COT values with the assistance of the auxiliary classifier ***C*** and the cross-attention module ***A***.

MT-HCCAR MODEL – 3.2

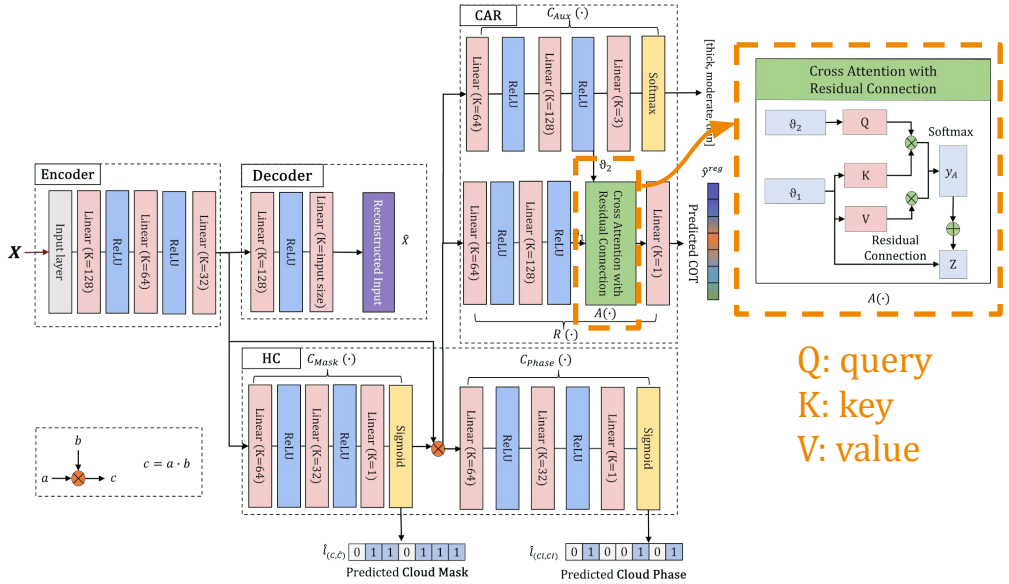


3.2 Auxiliary classifier (C)

- **D**iscretizes continuous COT values into coarse groups: [thick cloud, moderated cloud, thin cloud], based on COT values.
- Serving as a preprocessing step before regression to align pixels with similar characteristics in the feature space.



MT-HCCAR MODEL – 3.3

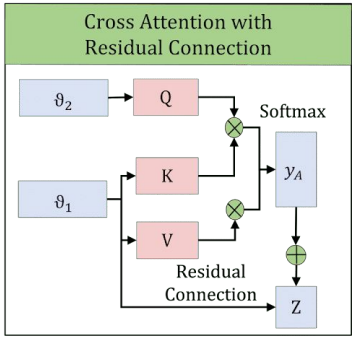


Q: query
K: key
V: value

3.3 Cross attention (A) with a residual connection

- Integrates features from both regression ***R*** and auxiliary classification ***C***.
- Facilitates deeper feature learning and more cohesive feature amalgamation.

MT-HCCAR MODEL – 3.3



$A(\cdot)$

$\boldsymbol{\vartheta_1}$ Features from Regression Network

$\boldsymbol{\vartheta_2}$ Features from Auxiliary Classifier

The cross attention with residual connection $\boldsymbol{A(\cdot)}$:

$$\boldsymbol{A}(\vartheta_1, \vartheta_2) = W_Z y_A(\vartheta_1, \vartheta_2) + \vartheta_1$$

W_Z : Model weights, learned through model training.

$y_A = softmax(QK^T)V$

Residual connection

$Q = W_Q \vartheta_2$ $K = W_K \vartheta_1$ $V = W_V \vartheta_1$

W_Q, W_K, W_V: Weight maps of query, key and value.

MODEL TRAINING

- The loss function $\mathcal{L}$ for training MT-HCCAR model

$$\mathcal{L} = \mathcal{L}_{HC} + \mathcal{L}_{CAR} + \mathcal{L}_{D} + \mathcal{L}_{Lasso}$$

Loss of HC

$$\mathcal{L}_{HC} = \mathcal{L}_{C_{Mask}} + \mathcal{L}_{C_{Phase}}$$

Loss of CAR

$$\mathcal{L}_{CAR} = \mathcal{L}_{R} + \mathcal{L}_{C_{Aux}}$$

Loss of the decoder

$$\mathcal{L}_{D} = \sum_{i=1}^{n} (X_i - \hat{X}_i)^2$$

Loss of Lasso regularization

$$\mathcal{L}_{Lasso} = \lambda \sum_{p=1}^{P} |\beta_p|^2$$

Loss of cloud masking

$$\mathcal{L}_{C_{Mask}} = BCE(y_{mask}, \hat{y}_{mask})$$

Loss of cloud phase classification

$$\mathcal{L}_{C_{Phase}} = BCE(\hat{y}_{mask} y_{phase}, \hat{y}_{mask} \hat{y}_{phase})$$

Loss of COT prediction

$$\mathcal{L}_{R} = L1(y_{COT}, \hat{y}_{COT})$$

Loss of auxiliary classification

$$\mathcal{L}_{C_{Aux}} = CE(y_{thickness}, \hat{y}_{thickness})$$

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EVALUATION METRICS

For classification tasks

- Cloud masking **accuracy**: binary accuracies between the two big categories [cloudy, cloud-free].
- Area under **precision-recall curve** $\boldsymbol{AU}\left(\overline{\boldsymbol{PRC}}\right)$ across all classes.

For the regression task

- Mean squared error (**MSE**)
- Coefficient of determination ($\boldsymbol{R^2}$)
- Fraction of pixels meeting PACE goals (**FMG**): the PACE Validation Plan defines a predictive model with satisfactory performance.
  - FMG should $\geq 65\%$, for cloudy pixels.

Please find the calculation in our paper.

EXPERIMENTS

- Baseline methods

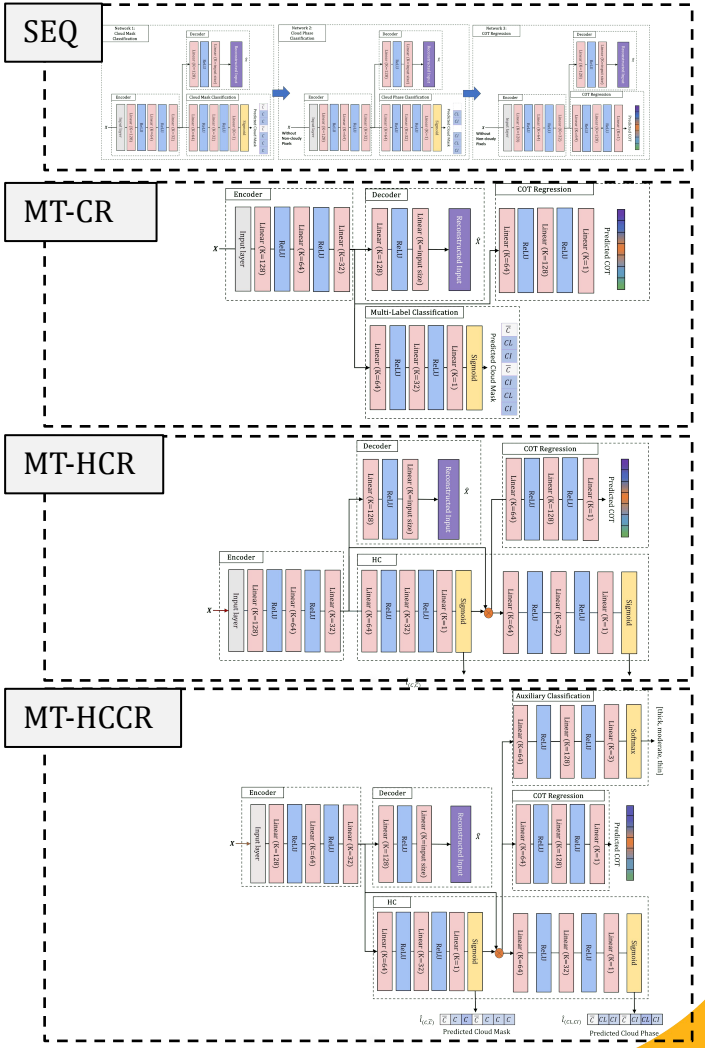
- **Chen et al. (2018)**: A Multi-layer Perceptron (MLP) based method proposed by Chen et al. [42] for cloud property retrieval. We use 1 hidden layer with 10 nodes, which are the same as the authors used.

- **Liu et al. (2022)**: A random forest (RF) based method implemented by Liu et al. [8] for cloud detection. We apply one RF to cloud masking and the same RF to cloud phase classification, respectively. The parameters involved are $ntrees = 100$ and $mdepth = 10$, which are the best parameters for RF found by the authors.

MODELS

- **Ablation studies**

Model	Multi-task learning?	Hierarchical classification?	Regression with auxiliary classifier?	Regression with attention?
SEQ	✗	✗	✗	✗
MT-CR	✓	✗	✗	✗
MT-HCR	✓	✓	✗	✗
MT-HCCR	✓	✓	✓	✗
MT-HCCAR	✓	✓	✓	✓



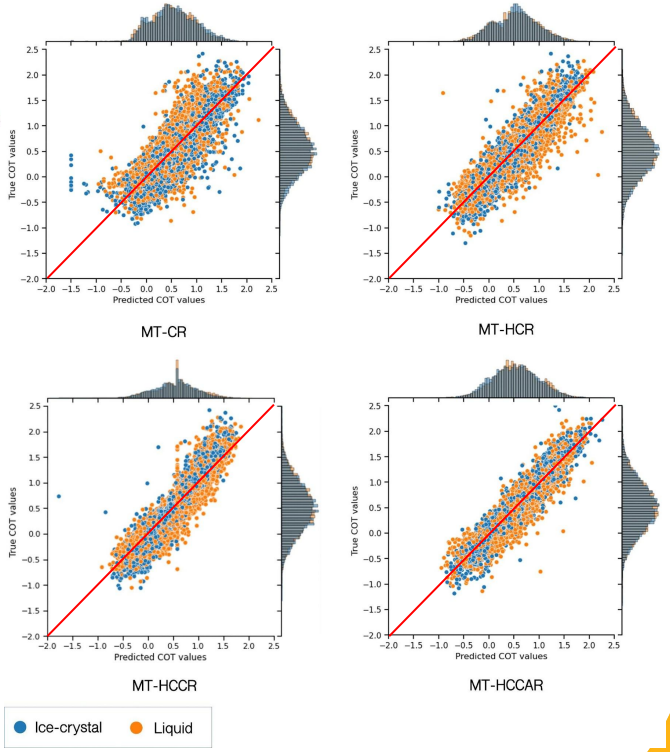
BASELINE COMPARISON & ABLATION STUDY

TABLE I
MODEL PERFORMANCE COMPARISON WITH BASELINE MODELS AND ABLATION STUDY - CLASSIFICATION RESULTS. MODELS ARE TRAINED BY OCI.

MODEL	ACC_{bi}	$AU(\overline{PRC})_w$	$AU(\overline{PRC})_c$			
			Cloudy	Cloud-free	Liquid cloud	Ice cloud
Liu et al.,2022 [8]	0.968	0.985	0.955	0.949	0.951	0.965
Chen et al., 2018 [42]	0.942	0.638	0.974	0.882	0.385	0.348
SEQ	0.963	0.750	0.997	0.978	0.618	0.658
MT-CR	0.965	0.955	0.925	0.925	0.954	0.967
MT-HCR	0.975	0.993	0.999	0.989	0.982	0.992
MT-HCCR	0.982	0.995	0.999	0.992	0.985	0.993
MT-HCCAR	**0.984**	**0.996**	**0.999**	**0.995**	**0.990**	**0.996**

TABLE II
MODEL PERFORMANCE COMPARISON WITH BASELINE MODELS AND ABLATION STUDY - REGRESSION RESULTS. MODELS ARE TRAINED BY OCI.

MODEL	MSE	MSE EACH LABEL		R^2	R^2 EACH LABEL		FMG	
		Liquid	Ice		Liquid	Ice	Liquid	Ice
Liu et al.,2022 [8]	-	-	-	-	-	-	-	-
Chen et al., 2018 [42]	0.185	0.193	0.178	0.235	0.247	0.212	43.92%	60.64%
SEQ	0.126	0.133	0.118	0.440	0.274	0.548	4.00%	41.11%
MT-CR	0.062	0.075	0.049	0.728	0.661	0.788	44.46%	68.21%
MT-HCR	0.038	0.045	0.030	0.843	0.810	0.874	65.93%	82.28%
MT-HCCR	0.031	0.035	0.028	0.869	0.854	0.884	66.38%	82.43%
MT-HCCAR	**0.026**	**0.032**	**0.021**	**0.891**	**0.869**	**0.914**	**67.96%**	**82.70%**



y-axis: true COT values. x-axis: predicted COT values.

MODEL SELECTION METHOD

MT-HCCAR seems to achieve the best performance, *but*…

Is MT-HCCAR showing significant improvement?

Is MT-HCCAR too complex?

MODEL SELECTION METHOD

Aim: to **select the optimal model** from {**MT-CR, MT-HCR, MT-HCCAR**} based on their performance over four metrics {Accuracy, $AU(\overline{PRC})$, MSE, R^2}, and over three simulated satellite datasets {**OCI, VIIRS, ABI**}.

- **STEP 1** : Conduct 10-fold cross validation
- **STEP 2** : Use one standard error rule (1 SE Rule)
- **STEP 3** : Calculate absolute performance score and 1SE score for each model to make the selection

MODEL SELECTION RESULTS

TABLE III
MODEL SELECTION USING K-FOLD CROSS-VALIDATION AND 1SE RULE.

DATASET	MODEL	$\mu(Metric)$	$SE(Metric)$	1SE RANGE	$P_{ab}^{\mathcal{M},\mathcal{D}}$	$P_{1SE}^{\mathcal{M},\mathcal{D}}$
		MEAN VALUE OF ACC_{bi} OVER 10 FOLDS				
OCI	MT-CR	0.969	7.608E-04	[0.969, 0.970]	-1.54%	0
	MT-HCR	0.984	**6.741E-04**	[0.983, 0.985]	-0.04%	**1**
	MT-HCCAR	**0.985**	1.057E-03	[0.983, 0.986]	**0.00%**	0
VIIRS	MT-CR	0.976	6.866E-04	[0.975, 0.976]	-1.39%	0
	MT-HCR	0.989	5.027E-04	[0.988, 0.989]	-0.04%	**1**
	MT-HCCAR	**0.989**	**4.961E-04**	[0.989, 0.990]	**0.00%**	0
ABI	MT-CR	0.973	1.229E-03	[0.972, 0.974]	-1.37%	0
	MT-HCR	0.986	**4.716E-04**	[0.986, 0.987]	-0.03%	**1**
	MT-HCCAR	**0.987**	6.180E-04	[0.986, 0.987]	**0.00%**	0
		MEAN VALUE OF $AU(PRC)_w$ OVER 10 FOLDS				
OCI	MT-CR	0.953	1.66E-03	[0.951, 0.954]	-4.29%	0
	MT-HCR	0.995	2.83E-04	[0.995, 0.995]	-0.04%	0
	MT-HCCAR	**0.996**	**2.08E-04**	[0.995, 0.996]	**0.00%**	**1**
VIIRS	MT-CR	0.939	3.17E-02	[0.908, 0.971]	-5.78%	0
	MT-HCR	0.996	1.88E-04	[0.996, 0.997]	-0.06%	0
	MT-HCCAR	**0.997**	**1.37E-04**	[0.997, 0.997]	**0.00%**	**1**
ABI	MT-CR	0.969	2.19E-03	[0.967, 0.971]	-2.75%	0
	MT-HCR	**0.996**	**1.62E-04**	[0.996, 0.996]	**0.00%**	**1**
	MT-HCCAR	**0.996**	1.67E-04	[0.996, 0.996]	-0.02%	0

DATASET	MODEL	$\mu(Metric)$	$SE(Metric)$	1SE RANGE	$P_{ab}^{\mathcal{M},\mathcal{D}}$	$P_{1SE}^{\mathcal{M},\mathcal{D}}$
		MEAN VALUE OF MSE OVER 10 FOLDS				
OCI	MT-CR	0.055	2.916E-03	[0.058, 0.052]	-100.00%	0
	MT-HCR	0.034	5.970E-04	[0.034, 0.035]	-25.27%	0
	MT-HCCAR	**0.027**	**4.910E-04**	[0.028, 0.027]	**0.00%**	**1**
VIIRS	MT-CR	0.043	2.075E-03	[0.045, 0.041]	-76.48%	0
	MT-HCR	0.030	**8.104E-04**	[0.031, 0.029]	-21.65%	0
	MT-HCCAR	**0.025**	1.282E-03	[0.026, 0.023]	**0.00%**	**1**
ABI	MT-CR	0.057	1.519E-03	[0.059, 0.056]	-78.86%	0
	MT-HCR	0.038	5.714E-04	[0.039, 0.038]	-19.48%	0
	MT-HCCAR	**0.032**	**3.121E-04**	**[0.032, 0.032]**	**0.00%**	**1**
		MEAN VALUE OF R^2 OVER 10 FOLDS				
OCI	MT-CR	0.758	1.24E-02	[0.770, 0.746]	-14.25%	0
	MT-HCR	0.854	2.55E-03	[0.857, 0.851]	-3.39%	0
	MT-HCCAR	**0.884**	**2.35E-03**	[0.882, 0.886]	**0.00%**	**1**
VIIRS	MT-CR	0.810	9.62E-03	[0.800, 0.820]	-9.72%	0
	MT-HCR	0.874	**3.59E-03**	[0.870, 0.878]	-2.58%	0
	MT-HCCAR	**0.897**	5.30E-03	[0.892, 0.902]	**0.00%**	**1**
ABI	MT-CR	0.751	7.44E-03	[0.743, 0.758]	-13.25%	0
	MT-HCR	0.840	2.20E-03	[0.838, 0.843]	-2.91%	0
	MT-HCCAR	**0.866**	**1.51E-03**	[0.864, 0.867]	**0.00%**	**1**

STEP 1: Conduct 10-fold cross validation

TABLE III
MODEL SELECTION USING K-FOLD CROSS-VALIDATION AND 1SE RULE.

DATASET	MODEL	$\mu(Metric)$	$SE(Metric)$	1SE RANGE	$P_{ab}^{\mathcal{M},\mathcal{D}}$	$P_{1SE}^{\mathcal{M},\mathcal{D}}$
		MEAN VALUE OF ACC_{bi} OVER 10 FOLDS				
OCI	MT-CR	0.969	7.608E-04	[0.969, 0.970]	-1.54%	0
	MT-HCR	0.984	**6.741E-04**	[0.983, 0.985]	-0.04%	**1**
	MT-HCCAR	**0.985**	1.057E-03	[0.983, 0.986]	**0.00%**	0
VIIRS	MT-CR	0.976	6.866E-04	[0.975, 0.976]	-1.39%	0
	MT-HCR	0.989	5.027E-04	[0.988, 0.989]	-0.04%	**1**
	MT-HCCAR	**0.989**	**4.961E-04**	[0.989, 0.990]	**0.00%**	0
ABI	MT-CR	0.973	1.229E-03	[0.972, 0.974]	-1.37%	0
	MT-HCR	0.986	**4.716E-04**	[0.986, 0.987]	-0.03%	**1**
	MT-HCCAR	**0.987**	6.180E-04	[0.986, 0.987]	**0.00%**	0
		MEAN VALUE OF $AU(\overline{PRC})_w$ OVER 10 FOLDS				
OCI	MT-CR	0.953	1.60E-03	[0.951, 0.954]	-4.29%	0
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	MT-HCR	0.034	5.970E-04	[0.034, 0.035]	-25.27%	0
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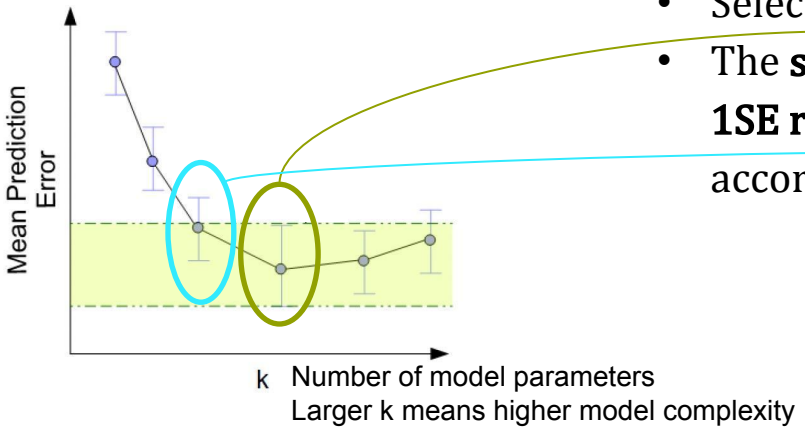
- Calculate the mean value for each metric over 10-folds:

$$\mu(Metric) = \frac{1}{K}\sum_{k=1}^{K=10} Metric^{k,\mathcal{M}^{\mathcal{D}}}$$

, where $Metric^k$ is $Metric$ for fold k of model $\mathcal{M}$ trained by dataset $\mathcal{D}$.

STEP 2: Use one standard-error rule (1 SE Rule)

- However, it is essential to consider the **trade-off between model complexity and absolute performance** .
- Therefore, in addition to the mean value, we also calculate the **1SE region** of each model.



1 SE Rule :

- Select ``**1SE region of the best** ''
- The **simplest model whose mean falls within 1SE region of the best is the optimal model** according to 1SE rule

STEP 2: Use one standard error rule (1 SE Rule)

Use 1-SE rule

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Standard error: $SE(Metric) = \frac{s}{\sqrt{K}}$,where s^2 is the variance, $K = 10$ is the number of folds

1SE range: $[\mu - SE, \mu + SE]$

STEP 3: Calculate Performance Scores

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STEP 3: Calculate Performance Scores

- **Absolute performance score** describes how different model performances are based on mean values of metrics

$$P_{ab}^{\mathcal{M}} = \sum_{D} \sum_{Metric} P_{ab}^{\mathcal{M},\mathcal{D}} = \sum_{D} \sum_{Metric} \frac{\mu(Metric)^{\mathcal{M}^{\mathcal{D}}} - \mu(Metric)^{\mathcal{M}_{\mu}^{\mathcal{D}\star}}}{\mu(Metric)^{\mathcal{M}_{\mu}^{\mathcal{D}\star}}} \quad (20)$$

, where components $P_{ab}^{\mathcal{M},\mathcal{D}}$ is the score for a specific model $\mathcal{M}$ applied to a specific dataset $\mathcal{D}$, $\mu(Metric)^{\mathcal{M}^{\mathcal{D}}}$ is the mean value of a specific metric $Metric$ of model $\mathcal{M}^{\mathcal{D}}$ applied to dataset $\mathcal{D}$, and $\mathcal{M}_{\mu}^{\mathcal{D}\star}$ is the model with the best mean value for dataset $\mathcal{D}$.

- **1SE rule performance score** describes how confident the model is chosen based on 1SE Rule.

$$P_{1SE}^{\mathcal{M}} = \sum_{D} \mathbf{w_m} \sum_{Metric} P_{1SE}^{\mathcal{M},\mathcal{D}} = \sum_{D} \sum_{Metric} \psi(\mu(Metric), \mu(\mathcal{M}, \mathcal{D}) \quad (21)$$

, where $\psi(\cdot)$ is a rule that if Model $\mathcal{M}$ trained by dataset $\mathcal{D}$ is selected by 1SE rule based on the mean value of a certain metric, then $\psi(\cdot)$ outputs 1, and otherwise 0. Weights $\mathbf{w_m} = [w_{Acc_{bi}}, w_{AU(\overline{PRC})_w}, w_{MSE}, w_{R^2}]$ define the significance of four metrics. We set $\mathbf{w_m} = [1, 1, 1, 1]$ in our experiments

MODEL SELECTION RESULTS

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MT-CR $P_{ab}^{MT-CR} = -4.12\%$ $P_{1SE}^{MT-CR} = 0$

MT-HCR $P_{ab}^{MT-HCR} = -0.21\%$ $P_{1SE}^{MT-HCR} = 4$

MT-HCCAR * $P_{ab}^{MT-HCCAR} = -0.02\%*$ $P_{1SE}^{MT-HCCAR} = 8*$

***The best trade-off balance between performance and complexity**

CONCLUSIONS

- In this study, we present MT-HCCAR, an end-to-end multi-task learning model for cloud property retrieval.
- Multi-task learning strategy enhances performance in both classification and regression tasks.
- Comparative analyses against two baseline methods and ablation studies prove the effectiveness of HC module and the CAR module.
- Moreover, our model selection approach, incorporating K-fold cross-validation, the 1SE rule, establishes MT-HCCAR's superior performance across 3 different datasets and 4 evaluation metrics, even when considering trade-off balance between model performance and complexity.

OPEN SOURCE

Code and data: *https://github.com/AI-4-atmosphere-remote-sensing/MT-HCCAR*

Please star us if this work helps :)

Thank You!

MT-HCCAR: Multi-Task Deep Learning Approach for Cloud Property Retrieval

Xingyan Li, Andrew M. Sayer, Ian T. Carroll, Xin Huang, Jianwu Wang
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